

CONTACT

Konstanz, Germany
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SKILLS

PROGRAMMING

Python (pandas, NumPy, scikit-learn)
R (tidyverse, ggplot2)
SQL, Bash, Stata, MATLAB
C++ (basic)

NLP & GENERATIVE AI

LLM evaluation and bias analysis
Prompt engineering
Transformer fine-tuning
Text classification
Word embeddings (Word2Vec, FastText)
PyTorch, Hugging Face Transformers, Gensim

MACHINE LEARNING & STATISTICS

Classification and regression
Statistical inference, hypothesis testing
Bootstrapping

DATA ENGINEERING & TOOLS

Relational databases, ETL
Web scraping, APIs
HPC clusters (Slurm)
Linux servers
PostgreSQL, Parquet, Git/Git LFS
Conda, unit testing, Jupyter
VS Code, Quarto, LaTeX

KATHARINA MAYER

Data Scientist specialising in NLP and Large Language Models
Interdisciplinary background in law and economics

EXPERIENCE

University of Konstanz
Prof. Dr. David Garcia
since 05/2026

Research Assistant

- **Detecting AI-generated legal text:** developed a Python pipeline (prompt design, quality checks, unit tests) that generates synthetic legal texts with open-source LLMs; building a benchmark dataset for detecting AI-generated text; runs via Slurm on an HPC cluster
- **Social media data:** scraped 1.48 million X/Twitter posts from German and British party and media accounts (2005–2022) with twscrape; resumable Python pipeline on a Linux server with monthly completeness checks

Deutscher Apotheker Verlag, Stuttgart
since 04/2024

Working Student, Rights and Licensing

- Digitisation and archiving of licence agreements
- Contract review and reconciliation of contract and system data

Parental leave
07/2021 – 03/2024

BRP Renaud und Partner mbB, Stuttgart
09/2020 – 10/2020

Legal Intern, Corporate Law and M&A

- Legal memos on corporate, civil and private international law, including GmbH redemption clauses and § 132 of the German Stock Corporation Act
- Assessed the applicability of the Rome I Regulation and the CISG to a German–Chinese sales contract; drafted contract documents

IBM Deutschland GmbH, Ehningen
08/2016 – 04/2017

Working Student and Intern, Hardware Sales

- Supported investment reviews and deal registrations; prepared decision documents for management

EDUCATION

University of Konstanz
10/2024 – present

M.Sc. Social and Economic Data Science

Grade	1.3 (provisional; German scale, 1.0 = best)
Focus	Deep learning, NLP and statistical modelling, applied to legal and social media data
Thesis	*Detecting Gender Bias in Judicial Language: A Projection-Based Analysis of Real and AI-Generated Opinions* (submitted 09/2026)

University of Konstanz
since 04/2019

Bachelor of Laws (LL.B.)

Grade	8.60 points (provisional; scale 0–18)
Focus	Labour and social security law
Thesis	*Use of the employer's resources for strike mobilisation* (13 points)

University of Konstanz
09/2014 – 10/2018

B.Sc. Economics

Grade	1.8
Focus	Applied economics
Thesis	*The Impact of Gender on Racial Discrimination in Grading* (grade 1.0)
Abroad	Charles University, Prague (2017/18)

LANGUAGES

German	native
English	fluent
Spanish, French	basic

INTERESTS

Cooking • Sports • Travel

PROJECTS

Master's thesis
2026

Gender Bias in Judicial Language

- **Data:** built a corpus of 1.14 million US appellate opinions in PostgreSQL, including attribution of opinions to judges via extracted names
- **Embeddings:** trained Word2Vec and FastText models and measured gender associations over time, across areas of law and by judge gender
- **LLMs:** counterfactual experiment with four LLMs (4,032 opinions on real cases, varying only the gender of a party) and analysis of bias in outcomes and language

Data Science Project
2026 • Grade 1.0

CourtListener Research Database

Manuscript under review at EPJ Data Science (submitted 11/2026)
Relational PostgreSQL database of US judges, their careers and their opinions, including author attribution and linkage to Federal Judicial Center case records
Zenodo DOI 10.5281/zenodo.23063945 • GitHub

Social Media Data Analysis
2025 • Grade 1.0

The Impact of AI-Generated Comments on Sentiment in YouTube Discussions

Collected 68,019 comments on climate videos via the YouTube Data API; detected AI-generated comments with a DeBERTa-based detector and classified sentiment with a fine-tuned RoBERTa model; analysis with multinomial logistic regression
GitHub

Group project
2025 • Grade 1.3

Gentrification Detection from Street View Images

Built a gentrification index for San Francisco from census data and classified Google Street View images with eight trained CNN variants